

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 5

Symmetry and Invariance: Why Some Operations Are Powerful

Assignment

Self-Study Curriculum

Choose a mathematical technique you use regularly in your work (regression, clustering, dimensionality reduction, a specific test, a simulation method). Write a one-page analysis of the symmetry it exploits. Structure it as:

1. What the technique does (one paragraph)
2. What the symmetry is, what transformation leaves what property unchanged (one paragraph)
3. When this symmetry breaks, what class of problem the technique fails on, explained in terms of the symmetry being absent (one paragraph)
4. What you would look for in a problem to know in advance whether the technique will work (one paragraph)